UNIVERSITY OF THE PUNJAB

B.S. in Computer Science Third Semester : Fall 2024

Roll No. [redacted]

Subject: Calculus and Analytical Geometry | **Paper: MS-251 N** | **Time: 3 Hrs.** | **Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Define the following: **(6x5=30)**

(i) Evaluate $\lim_{x\to 1}\frac{x-1}{\sqrt{x+3}-2}$

(ii) Find the slope of the graph of $g(x)=\frac{8}{x^2}$ at $(2,2)$. Also find an equation of the tangent line to the graph.

(iii) Find the derivative of the function $f(x)=(3-x^2)(x^2-x+1)$.

(iv) Find the absolute extrema of the function $f(x)=\sqrt{4-x^2}$ in interval $[-2,1]$.

(v) Solve $\int_9^4 \frac{1-\sqrt{u}}{\sqrt{u}}du$.

(vi) Evaluate $\int(2x^2+e^{2x})dx$.

Q.2. Answer the following questions. **(6x5=30)**

(a) For what value of b, the following function is continuous?

$$f(x)=\begin{cases} x & x<-2 \\ bx^2 & x\geq -23 \end{cases}$$

(b) Prove that the function $\frac{|x|}{x}$ is not continuous at $x=0$.

(c) Find $\frac{dr}{d\theta}$ for $r=\theta\sin\theta+\cos\theta$.

(d) Use implicit differentiation to find $\frac{dy}{dx}$ in $x^2y+y^2x=6$.

(e) Find the value of c that satisfies the equation $f'(c)=\frac{f(b)-f(a)}{b-a}$ for the function $f(x)=x+\frac{1}{x}$, $[\frac{1}{2},2]$.

(f) Integrate by substitution $\int\sqrt{x}\sin^2(x^{\frac{3}{2}-1})dx$.